Unclipped three-variable gradient descent: certified capture and critical cusp selection

Abstract

For an explicit rational initial point, we certify a complete 100000-step prefix of raw simultaneous gradient descent and prove convergence using a positively invariant analytic tube. The orbit genuinely crosses the clipping threshold and never has an exactly zero noise coordinate. Separately, a discrete all-time argument establishes a one-parameter critical selection law with an analytic quadratic remainder.

Keywords: unclipped gradient descent; invariant capture tube; critical cusp selection; minimizer selection; validated finite prefix

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1 The exact unprojected map and its source

Consider ordinary simultaneous gradient descent with stepsize $1/30$ on

$$f(a,b,c) = 50a^2b^2 + (ac-1)^2/200.$$

Writing $u = 10/3$ and $h = 1/3000$, its map is

$$a^+ = a - uab^2 - hc(ac-1), \quad b^+ = (1-ua^2)b, \quad c^+ = c - ha(ac-1). \tag{1}$$

All right sides use the old state; there is no projection, cutoff or underflow convention. The final NeurIPS 2025 source [1] defines a projected theoretical algorithm in Section 3 and explicitly clips the noise coordinate in Appendix E. Appendix G instead experiments with the unconstrained problem. Our rational tuple is selected from the printed initialization set, not asserted to be a figure seed. The source's final version and arXiv v1 differ; the attribution here is to the final version. We do not assert that 10000 meets an unspecified "sufficiently large" feature-ratio constant.

Theorem 1. *For $x_0 = (13/20, 3/2000, 7/10)$, the exact orbit of (1) is bounded and converges to a global minimizer $(a_\infty, 0, c_\infty)$, with $a_\infty c_\infty = 1$. At every finite index, $|a_k|, |b_k|, |c_k| < 2$ and $b_k \neq 0$. Moreover*

$$0.7745766 < a_\infty < 0.7745784, \qquad 1.29102586 < c_\infty < 1.29102714.$$

The first index with $|b_k| > 1/20$ is $k = 2556$.

The theorem combines an outward-rounded finite interval calculation with an analytic infinite tail. A small open neighborhood of this initial tuple also converges to the minimizer manifold. No useful explicit basin radius or global removal-of-clipping theorem is claimed.

2 Finite interval certificate

The supplied Python certificate represents intervals by integer pairs (l, u) meaning $[l/S, u/S]$, $S = 10^{2000}$. Products take extrema of the four endpoint products and round outwards by integer floor and ceiling. Division by positive integers is rounded outwards; squaring treats a zero-crossing interval separately. Addition and negation are exact. Every simultaneous update uses the preceding a, b, c and residual. Induction therefore encloses every exact state through $N = 100000$.

The program checks all states for $|a|, |b|, |c| < 2$ and checks all noise magnitudes before the first crossing. It certifies

$$0.774577 \le a_N \le 0.774578, \qquad |b_N| < 0.000002,$$
$$1.291026 \le c_N \le 1.291027, \qquad |a_N c_N - 1| < 0.0000001.$$

The first crossing is 2556, with

$$0.059349664513207805 \le b_{2556} \le 0.059349664513207806.$$

All comparisons are integer comparisons. A separate C++ interval implementation using exact GMP integers on the binary grid 2^{-7000} certifies the same complete prefix; its source and execution log are included. Checkpoint displays are coarsened outward and are not substituted for the propagated state. In particular, a displayed zero endpoint for very small b does not set the coordinate to zero. The finite calculation relies on the supplied arithmetic implementations; it is not a formal proof-assistant certificate.

3 A positively invariant capture tube

Put $r = ac - 1$ and choose exact rational constants

$$a_- = 0.77457, \quad a_+ = 0.77459, \quad c_- = 1.29100, \quad c_+ = 1.29105,$$
$$B = 0.000002, \quad R_0 = 0.0000001, \quad q = 0.99997, \quad p = 0.9993, \quad D = 4.$$

Let O consist of $a \in [a_-, a_+]$, $c \in [c_-, c_+]$, $|b| \le B$, $|r| \le R_0$. Expansion of (1) gives

$$r^+ = Kr - uacb^2, \qquad K = 1 - h(a^2 + c^2) + uha^2b^2 + h^2acr.$$

Exact rational endpoint bounds, reproduced by the tail checker, give on O

$$0 < K < p, \qquad |1 - ua^2| < q, \qquad uac < D, \qquad pR_0 + DB^2 < R_0.$$

Thus $|b^+| \le q|b|$ and $|r^+| \le p|r| + Db^2$. Define globally the budget functions

$$S_b = \frac{b^2}{1 - q^2}, \qquad V = \frac{|r| + DS_b}{1 - p}, \qquad A_b = ua_+S_b + hc_+V, \qquad C_b = ha_+V.$$

The trimmed tube T is the set of points in O satisfying

$$a_- + A_b \le a \le a_+ - A_b, \qquad c_- + C_b \le c \le c_+ - C_b.$$

Lemma 1. *T is positively invariant, and every orbit starting in T converges to a single point of $\{(a, 0, c) : ac = 1\}$.*

Proof. For a current point in T, the scalar estimates, before any assumption on its image, imply

$$S_b(x^+) \le S_b(x) - b^2, \qquad V(x^+) \le V(x) - |r|.$$

The second follows by substituting the first into the definition of V and canceling the Db^2 term. The coordinate updates imply

$$|a^+ - a| \le ua_+b^2 + hc_+|r| \le A_b(x) - A_b(x^+), \quad |c^+ - c| \le ha_+|r| \le C_b(x) - C_b(x^+).$$

Hence $a^+ - A_b(x^+) \ge a_-$ and $a^+ + A_b(x^+) \le a_+$, with the analogous c bounds. The scalar contractions also give $|b^+| \le B$, $|r^+| \le R_0$. This proves actual invariance rather than presupposing it from a drift bound. Telescoping gives $\sum b_k^2 \le S_b(x_0)$ and $\sum |r_k| \le V(x_0)$, and the two total coordinate variations are bounded by $A_b(x_0)$ and $C_b(x_0)$. Thus a, c are Cauchy, $b \to 0$, and $r \to 0$; the limit has product one. ∎

Proof of Theorem 1. The terminal interval is inside T. Indeed the following rational worst-case budgets hold throughout $|b| \le B$, $|r| \le R_0$:

$$S_b \le \frac{1}{14999775}, \qquad V \le \frac{2199991}{4199937000},$$
$$A_b \le \frac{100183007611}{2519962200000000000} < 4 \cdot 10^{-7},$$
$$C_b \le \frac{170409102869}{12599811000000000000} < 1.4 \cdot 10^{-7}.$$

Every terminal a, c endpoint is strictly farther from the corresponding boundary of O than these budgets. The preceding lemma gives convergence and the stated limiting intervals after adding and subtracting the variation budgets. Combining the finite prefix and tube proves global boundedness of this orbit.

Every finite iterate is rational. From nonzero b_k, an exact zero at the next step would require $a_k^2 = 3/10$, impossible for rational a_k. Induction proves $b_k \ne 0$. The finite calculation establishes the crossing assertion. The clipped and raw maps agree through state 2555 but differ at state 2556, where clipping would give 0.05 rather than the enclosed raw value. Thus clipping is genuinely relevant to the selected orbit. ∎

Because F^N for the polynomial map F is continuous and terminal entry is strict, an open neighborhood of the rational tuple maps into the interior of T. All those orbits converge to the minimizer manifold. This existence statement does not give a uniform theorem over the source's initialization set.

4 Critical one-parameter selection

The same map has a second, entirely analytic conclusion. Let $A = \sqrt{3/5}$, $C = A^{-1}$ and start at (A, ε, C), a distinct initialization family chosen here.

Theorem 2. *There exists $\varepsilon_0 > 0$ such that, whenever $0 < |\varepsilon| < \varepsilon_0$, the entire orbit remains within $O(|\varepsilon|)$ of $(A, 0, C)$, has nonzero alternating-sign b_n at every finite index, and converges to a global minimizer. The selected coordinates satisfy*

$$a_\infty = A - \frac{3}{2\sqrt{17}}|\varepsilon| + O(\varepsilon^2), \qquad c_\infty = C + \frac{5}{2\sqrt{17}}|\varepsilon| + O(\varepsilon^2).$$

No explicit usable value of ε_0 or uniform rate as $\varepsilon \to 0$ is claimed.

Proof of Theorem 2.

4.1 Exact coordinates and identities

Write $r = ac - 1$, $v = |b|$, and define $p > 0$ uniquely by

$$p^2 - p^{-2} = a^2 - c^2.$$

Near the positive point (A, C), (p, r) is a smooth coordinate system: the derivative of $d(p) = p^2 - p^{-2}$ is $2p + 2p^{-3} > 0$, and the Jacobian of $(a, c) \mapsto (a^2 - c^2, ac - 1)$ is $2(a^2 + c^2) > 0$. In particular,

$$a = p + O(r), \qquad c = p^{-1} + O(r),$$

uniformly in a fixed small neighborhood. Set $s = p - A$. Our initial conditions are exactly $s_0 = r_0 = 0$, $v_0 = |\varepsilon|$.

Direct expansion of the simultaneous map gives the exact identities

$$r^+ = Q(a, b, c, r)r - uacv^2, \quad Q = 1 - h(a^2 + c^2) + uha^2v^2 + h^2acr, \tag{S1}$$

$$(a^+)^2 - (c^+)^2 - (a^2 - c^2) = -2ua^2v^2 + u^2a^2v^4 + 2uhacrv^2 - h^2(a^2 - c^2)r^2. \tag{S2}$$

The linear residual term cancels exactly in (S2). This is the reason to use p instead of a as the tangent coordinate.

4.2 A genuine invariant residual cone

At the base point, $Q_0 = 1 - h(A^2 + C^2)$ lies strictly between zero and one, while $ua^2 - 1 = 1$. Choose fixed constants

$$Q_0 < q < 1, \qquad \sqrt{q} < m < 1.$$

By continuity there is a neighborhood of the base point on which

$$0 < Q \le q, \quad ua^2 - 1 \ge m, \quad 0 < uac \le D$$

for a finite constant D, and a, c, a^+, c^+ are positive. Fix $L \geq D/(m^2 - q)$. Shrink the coordinate neighborhood as needed. If its current point satisfies

$$-Lv^2 \leq r \leq 0, \tag{S3}$$

then (S1) and $v^+ = (ua^2 - 1)v$ imply

$$r^+ \leq 0, \quad |r^+| \leq (qL + D)v^2 \leq Lm^2v^2 \leq L(v^+)^2. \tag{S4}$$

Thus the cone propagates at each step for which the current point is in the neighborhood. This assertion alone does not assume that the orbit stays there. The all-time capture argument below supplies that missing capture argument.

4.3 Uniform reduced-map estimates, with stable effects retained

On (S3), the coordinate estimates give $a = p + O(v^2)$. Applying Taylor's theorem to the smooth inverse of d in (S2) yields

$$s^+ - s = -k(p)v^2 + O(v^4), \qquad k(p) = \frac{2up^2}{2p + 2p^{-3}} = \frac{up^3}{p^2 + p^{-2}}. \tag{S5}$$

Every error here is uniform in a fixed sufficiently small neighborhood; its constant may depend on the fixed L. In particular, the r^2 term is bounded by L^2v^4, so stable-coordinate effects have not been dropped.

Let

$$K = k(A) = \frac{uA^3}{A^2 + C^2} > 0, \qquad \kappa = 2uA > 0.$$

After shrinking the neighborhood, there are positive finite constants k_*, k^*, M such that

$$-k^*v^2 \leq s^+ - s \leq -k_*v^2, \qquad k_* = K/2 \text{ is admissible,} \tag{S6}$$

and

$$s^+ - s = -Kv^2 + O(|s|v^2 + v^4), \quad v^+ = v\{1 + \kappa s + O(s^2 + v^2)\}. \tag{S7}$$

These estimates follow directly from (S5) and the exact multiplier $ua^2 - 1$, which stays positive. They are one-step estimates, so there is no flip-mode parity approximation.

Define

$$\lambda = K/\kappa = \frac{A^2}{2(A^2 + C^2)} = \frac{9}{68}, \qquad H = s^2 + \lambda v^2.$$

Expansion of (S7) gives the crucial uniform bound

$$|H^+ - H| \le MHv^2. \tag{S8}$$

For completeness, the only degree-three terms in the difference are $-2Ksv^2$ from $(s^+)^2 - s^2$ and $2\lambda\kappa sv^2$ from $\lambda[(v^+)^2 - v^2]$; these cancel. All remaining terms are bounded by a constant times $(s^2+v^2)v^2$, hence by MHv^2. In particular no unsupported claim that H is exactly invariant or monotone is made.

4.4 All-time capture: noncircular induction

Put $e = |\varepsilon|$, so $H_0 = \lambda e^2$. Choose $e_0 > 0$ sufficiently small that

- $H \le 2\lambda e^2$, $-Lv^2 \le r \le 0$, and $s \le 0$ imply membership in the fixed neighborhood where (S4)–(S8) hold, for $e < e_0$;
- the corresponding one-step image has positive a^+, c^+, as ensured when the neighborhood was chosen;
- with
$$B_e = \sqrt{2\lambda}\, e + 2k^* e^2,$$
one has $\exp(MB_e/k_*) \le 3/2$.

All three requirements hold for sufficiently small e_0, because $B_e \to 0$.

We prove by induction that at every time

$$s_n \le 0, \quad -Lv_n^2 \le r_n \le 0, \quad H_n \le 2H_0. \tag{S9}$$

They hold initially. Suppose they hold through time n. The previous-point estimates are therefore legitimate through the update to $n+1$. Equation (S4) propagates the cone, and (S6) propagates $s \le 0$. Moreover, using only the inductive bounds at time n,

$$-s_{n+1} \le |s_n| + k^* v_n^2 \le B_e.$$

Telescoping (S6) from the exact value $s_0 = 0$, including this last update, yields

$$\sum_{j=0}^{n} v_j^2 \le \frac{-s_{n+1}}{k_*} \le B_e/k_*.$$

By (S8), applied only at the already justified previous points,

$$H_{n+1} \le H_0 \prod_{j=0}^{n}(1 + Mv_j^2) \le H_0 \exp\left(M \sum_{j=0}^{n} v_j^2\right) \le \tfrac{3}{2} H_0 < 2H_0.$$

This closes the induction. In particular it proves actual local capture, rather than assuming a trapped orbit while estimating its total drift.

Consequently $|s_n| \le \sqrt{2\lambda}e$, $v_n \le \sqrt{2}e$, and $|r_n| \le 2Le^2$ for all n. The original variables satisfy the claimed $O(e)$ neighborhood bound.

There is also an exact amplitude monotonicity consequence. At fixed $a^2-c^2 = p^2-p^{-2}$, the positive solution satisfies

$$a^2 = \frac{d(p) + \sqrt{d(p)^2 + 4(1+r)^2}}{2}.$$

For $-1 < r \le 0$ this implies $a \le p$. Since capture proved $p_n \le A$, one has $m \le ua_n^2 - 1 \le 1$, and hence $v_{n+1} \le v_n \le e$. This strengthening is not needed in the bootstrap above.

4.5 Infinite tail and point selection

By (S6) and (S9), s_n is decreasing and bounded below, so it converges to a limit $s_\infty \le 0$. More sharply,

$$\sum_{n=0}^{\infty} v_n^2 \le \frac{-s_\infty}{k_*} \le \frac{\sqrt{2\lambda}}{k_*}e. \tag{S10}$$

Thus $v_n \to 0$, and the cone implies $r_n \to 0$. The coordinate inverse now proves convergence of the actual three-variable orbit:

$$a_n \to A + s_\infty, \quad b_n \to 0, \quad c_n \to (A + s_\infty)^{-1}.$$

This is a global minimizer because both nonnegative terms of the stated objective vanish there. The multiplier stays negative and bounded away from zero, so $b_n \neq 0$ and its sign alternates at every finite step.

The full accumulated error in (S8), not just a fixed-horizon error, is controlled using (S9)–(S10):

$$|H_\infty - H_0| \le \sum_{n=0}^{\infty} |H_{n+1} - H_n| \le 2M\lambda e^2 \sum_{n=0}^{\infty} v_n^2 = O(e^3).$$

Since $v_n \to 0$, this gives

$$s_\infty^2 = \lambda e^2 + O(e^3).$$

The negative square-root branch is forced by (S6): already $s_1 \le -k_* e^2 < 0$, and s_n never increases. Hence

$$s_\infty = -\sqrt{\lambda}\, e + O(e^2) = -\frac{3}{2\sqrt{17}}e + O(e^2).$$

Finally expand $(A + s_\infty)^{-1}$ and use $A^{-2} = 5/3$. This proves both asserted constants and the $O(\varepsilon^2)$ errors.

For each fixed nonzero sufficiently small ε, the limit has $0 < a_\infty < A$. Therefore the limiting absolute b-multiplier is strictly below one. By convergence, b_n eventually decays geometrically; (S1) then gives geometric residual decay, and summable geometric coordinate increments give a genuine attracting tail. No rate uniform as $\varepsilon \to 0$ is claimed. ∎

5 Reproduction and limitations

Run the certificate scripts as described in the accompanying README. The independent prefix calculation uses GMP; the Python calculation uses integer arithmetic, and the tail checker uses exact rational arithmetic. These certify only the finite component of Theorem 1. Theorem 2 has no numerical-prefix dependency. Neither theorem proves capture on the transversely unstable side of the critical minimizer or a global clipping-removal theorem.

The final source PDF has SHA-256

`e34e316e2714b580dff222ee7885cf7eb4`
`395fb34bb147ef8a0903574ad3f5dc.`

Only the specified one-parameter family is included. No two-parameter selection law is asserted.

References

[1] Liu, Zhang, Du and Zhao. *A Minimalist Example of Edge-of-Stability and Progressive Sharpening.* Advances in Neural Information Processing Systems, 2025, final proceedings version. `https://papers.nips.cc/paper_files/paper/2025/file/15eac388458e8a577f96edf7a40d0cbc-Paper-Conference.pdf`.